$$\frac{20ft}{5s} = 4 \, ft/s \qquad \frac{0}{0}$$

x: position

$$\Delta x \qquad \frac{\Delta x}{\Delta t}$$

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x) = 3x - 2$$

$$\lim_{h \to 0} \frac{(3(x+h) - 2) - (3x - 2)}{h}$$

$$\frac{3x + 3h - 2 - (3x - 2)}{h}$$

$$\frac{3h}{h} = 3$$

$f(x) = (x^2 + 3x + 1)$

$$\lim_{h \to 0} \frac{(x+h)^2 + 3(x+h) + 1 - (x^2 + 3x + 1)}{h}$$

$$\frac{\cancel{x^2} + 2xh + h^2 + \cancel{3x} + 3h + \cancel{1} - \cancel{x^2} - \cancel{3x} - \cancel{1}}{h}$$

$2x + h + 3$

$f'(x) = 2x + 3$

$\frac{d}{dx} f(x) = 2x + 3$

$x^3 \rightarrow 3x^2$

x^3

$x^5 \rightarrow 5x^4$

$x^{-1} \rightarrow -x^{-2}$

$\sqrt{x}$ $\quad x^{\frac{1}{2}} \rightarrow \frac{1}{2} x^{-\frac{1}{2}}$

$\frac{d}{dx}(7x^4 - 9x^3 + 2x + 5)$

$= 28x^3 - 27x^2 + 2 + 0$

$$(f(x)+g(x))' = f'(x) + g'(x)$$

$$(x^2-1)' = 2x$$

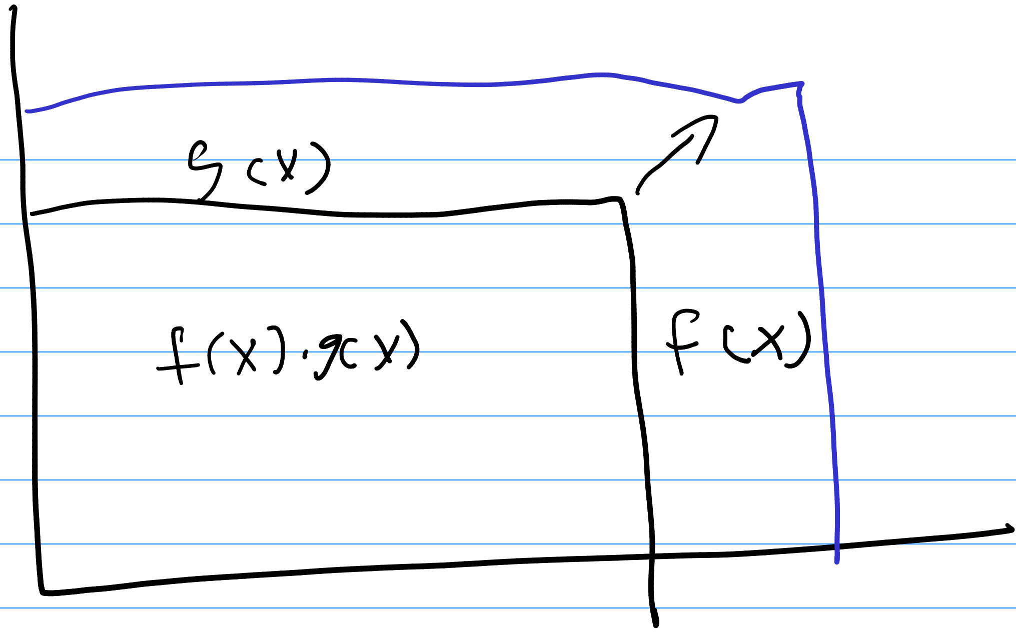
$$((x+1)(x-1))' = 2x$$

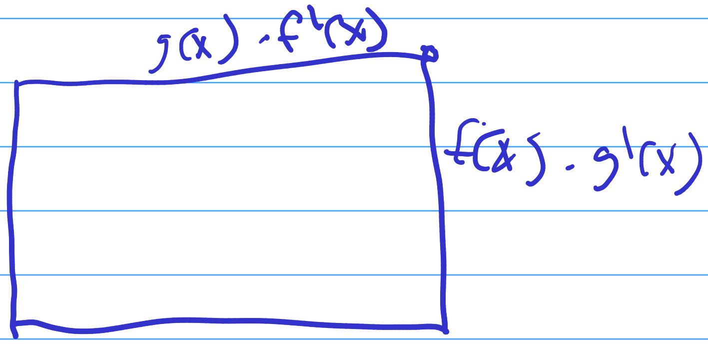
$$(x+1)' \cdot (x-1)'$$

$$1 \cdot 1 = 1 \neq 2x$$

$$(a+b)(c+d)$$

	a	b
c	ac	bc
d	ad	bd





$$[f(x)\cdot g(x)]' = f(x)\cdot g'(x) + g(x)\,f'(x)$$

Product rule

$$(x-1)(x+1) \qquad (x^2-1)' = 2x$$

$$\underset{f(x)}{\uparrow} \qquad \underset{g(x)}{\uparrow}$$

$$(x-1)\cdot 1 + (x+1)\cdot 1$$

$$x-1+x+1 = 2x$$

$$(3x-2)(4x^2+3x)$$

$$\underset{f(x)}{\uparrow} \qquad \underset{g(x)}{\uparrow}$$

$$\underset{f}{(3x-2)}\,\underset{g'}{(8x+3)} + \underset{g}{(4x^2+3x)}\,\underset{f'}{(3)}$$

$$\frac{f(x)}{g(x)} = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$$

$$y = \frac{(x^2 - 1) \leftarrow f}{(x+1) \leftarrow g} \qquad x-1$$

$$y' = \frac{(x+1)(2x) - (x^2-1)(1)}{(x+1)(x+1)}$$

$$\frac{\cancel{(x+1)}(2x) - \cancel{(x+1)}(x-1)}{\cancel{(x+1)}(x+1)}$$

$$\frac{2x - (x-1)}{x+1} = \frac{x+1}{x+1} = 1$$

$$y = \frac{-x^2 + 4x - 1.5}{(3x-2)}$$

$$\frac{(3x-2)(-2x+4) - (-x^2+4x-1.5)(3)}{(3x-2)^2}$$

$$e = 2.71828$$

$$\frac{d}{dx} e^x = e^x$$

$\ln(x)$ $\ln x$

$\log_e x$

$\ln(2) = 0.69\ldots$

$e^{0.69\ldots} = 2$

$$(\ln x)' = \frac{1}{x}$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$y = \ldots$

$\frac{dy}{dx}$ y'

$$\frac{d}{dx} f(g(x)) \qquad f'(g(x)) \cdot g'(x)$$

$$y = f(\underbrace{g(x)}_{u}) \qquad \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}$$

$$y = \sin(\ln(x))$$

$$y' \cos(\ln(x)) \cdot \left(\frac{1}{x}\right)$$

$$\frac{\cos(\ln(x))}{x}$$

$$y = e^{(\sin(x^2))}$$

$$y' = \left(e^{(\sin(x^2))}\right)\left((\cos(x^2)\right)(2x)$$

33. $f(x) = \sin(3x+4)\cos(5-2x)$

tgglin.github.io/Weston2026

$$\sin(3x+4)\cdot\left(-\sin(5-2x)(-2)\right)$$
$$+\cos(5-2x)\cdot\left(\cos(3x+4)\cdot 3\right)$$

$$\frac{f(x)}{g(x)} = f(x)\cdot\frac{1}{g(x)} = f(x)\,(g(x))^{-1}$$

$$f(x)\left(-(g(x))^{-2}\cdot g'(x)\right) + (g(x))^{-1}(f'(x))$$

$$\frac{f(x)g'(x)}{-g(x)^2} + \frac{f'(x)}{g(x)}$$

$$-\frac{f(x)g'(x)}{g(x^2)}+\frac{f'(x)(g(x))}{g(x)^2}$$

$$\frac{f'(x)(g(x))-g'(x)f(x)}{(g(x))^2}$$